L'Oréal Warehouse Management System Documentation

- **Translated from French to English**

I. Revision History

Version	Description	Date	Author
1.0	Initial Version	10/10/2025	Rémi BAUSSAND
1.1	Added ASN report Added location descriptions Added LO_PRE zone for reserve locations Various corrections	14/10/2025	Rémi BAUSSAND

II. Context

Activity Name: MACASLEA1
Client: L'ORÉAL
Area: 1 cell of 5000 m^2 + 1500 m^2 (for copacking activity)

The AGL Morocco activity groups 4 different divisions within L'Oréal:

Division	Full Name	SAP Code (storage location)	Speed Quality Code	Description
10	LPD – Luxury Products Division	RL01	RL01	Luxury products
20	CPD – Consumer Products Division	RC01	RC01	Consumer products
30	PPD – Professional Products Division	RP01	RP01	Professional products
40	DPD – Dermatologic Products Division	RA01	RA01	Dermatological products

The divisions are a structuring element of the activity, as products are not stored in the same locations and order typologies vary considerably.

For example, CPD (consumer) handles large volumes, often full pallets, while LPD (luxury) manages smaller volumes, with a high proportion of mezzanine picking.

Within each division, some products are distinguished:

- Either by their level of danger (aerosols, risk levels)
- Or by their specific use (PLV – point-of-sale advertising)

Some items have a lot number with an expiration date (DLUO) automatically calculated from decoding the lot.

Others require an additional scan during preparation to ensure tracking of the AD Code – equivalent to a serial number – particularly for LPD and PPD divisions, for traceability and anti-fraud purposes.

These specificities lead to the coexistence of multiple storage and management rules. These will be detailed in the corresponding section. Their implementation required several specific developments, including:

- Entry of logistic data upon receipt
- Batch decoding
- AD Code tracking

Finally, the activity includes a copacking component, i.e., the creation of commercial bundles grouping multiple products.

This flow relies on:

- Preparation of components via DO Copacking
- Then, after assembly, receipt of the BOM ASN

This activity is generally performed under a specific quality code "RLSC".

There are also two other types of ASN:

- INB (international)
- PPP (local product)

To conclude, the activity also includes management of customer returns, divided into two categories:

- Cancellation (with reference)

- Return (without reference)

Cancellation can be total or partial. A specific process allows flagging the order to be canceled (status "shipped" mandatory). This flag automatically generates an expected receipt and corresponding reception.

Returns, on the other hand, are performed manually, via creation of a reception without expected receipt.

- **III. Layout**

Locations

Location nomenclature: LO001A001

All L'Oréal locations begin with the prefix "LO".
The next three digits identify the aisle.
The following letter indicates the level, with letter A corresponding to ground level.
In the pallet rack picking area, an intermediate beam was created between levels A and B to store cartons in dynamic picking.
This beam is identified by the letter C. Ground level A is generally dedicated picking.
The last three digits of the location code determine the lateral position on the beam.

The warehouse is divided into two main zones:

- A pallet racking zone, corresponding to aisles 01 to 14
- A mezzanine zone, dedicated to shelf storage (cartons only), covering locations 300 to 500 in dynamic picking.

Aisle	First_Location	Last_Location	Type
1	LO001A001	LO001H030	Rack
2	LO002A001	LO002H072	Rack
3	LO003A001	LO003H072	Rack
4	LO004A001	LO004H072	Rack
5	LO005A001	LO005H072	Rack

Aisle	First_Location	Last_Location	Type
6	LO006A001	LO006H072	Rack
7	LO007A001	LO007F072	Rack
8	LO008A001	LO008F072	Rack
9	LO009A001	LO009F072	Rack
10	LO010A001	LO010F072	Rack
11	LO011A001	LO011H012	Rack
12	LO012A001	LO012H012	Rack
13	LO013A001	LO013H012	Rack
14	LO014A001	LO014H012	Rack
15	LO015A001	LO015I071	Rack
16	LO016A001	LO016F032	Rack
17	LO017A001	LO017F033	Rack Aerosol
18	LO018A001	LO018F033	Rack Aerosol
20	LO020A001	LO020A108	Mass
21	LO021A001	LO021A108	Mass
30	LO030A001	LO030E024	Rack
31	LO031A001	LO031E028	Rack
32	LO032A001	LO032E028	Rack
33	LO033A001	LO033E024	Rack
40	LO040A001	LO040B108	Mass
41	LO041A001	LO041B108	Mass

Aisle	First_Location	Last_Location	Type
42	LO042A001	LO042B108	Mass
43	LO043A001	LO043B108	Mass
101	LO101A001	LO101E022	Dynamic mezzanine
102	LO102A001	LO102E022	Dynamic mezzanine
103	LO103A001	LO103E022	Dynamic mezzanine
104	LO104A001	LO104E019	Dynamic mezzanine
105	LO105A001	LO105E019	Dynamic mezzanine
106	LO106A001	LO106E022	Dynamic mezzanine
107	LO107A001	LO107E022	Dynamic mezzanine
108	LO108A001	LO108E022	Dynamic mezzanine
109	LO109A001	LO109E088	Dynamic mezzanine
110	LO110A001	LO110E092	Dynamic mezzanine
111	LO111A001	LO111E028	Dynamic mezzanine
112	LO112A001	LO112E036	Dynamic mezzanine
113	LO113A001	LO113E036	Dynamic mezzanine
114	LO114A001	LO114E036	Dynamic mezzanine
201	LO201A001	LO201E066	Dynamic mezzanine
202	LO202A001	LO202E066	Dynamic mezzanine
203	LO203A001	LO203E066	Dynamic mezzanine
204	LO204A001	LO204E057	Dynamic mezzanine
205	LO205A004	LO205E057	Dynamic mezzanine

Aisle	First_Location	Last_Location	Type
206	LO206A001	LO206E066	Dynamic mezzanine
207	LO207A001	LO207E066	Dynamic mezzanine
208	LO208A001	LO208E066	Dynamic mezzanine
209	LO209A001	LO209E066	Dynamic mezzanine
210	LO210A001	LO210E069	Dynamic mezzanine
211	LO211A001	LO211E021	Dynamic mezzanine
212	LO212A001	LO212E032	Dynamic mezzanine
213	LO213A001	LO213E028	Dynamic mezzanine
214	LO214A001	LO214E028	Dynamic mezzanine
301	LO301A001	LO301E115	Dynamic mezzanine
302	LO302A006	LO302E110	Dynamic mezzanine
303	LO303A006	LO303E110	Dynamic mezzanine
304	LO304A006	LO304E095	Dynamic mezzanine
305	LO305A006	LO305E095	Dynamic mezzanine
306	LO306A006	LO306E110	Dynamic mezzanine
307	LO307A006	LO307E110	Dynamic mezzanine
308	LO308A006	LO308E110	Dynamic mezzanine
309	LO309A006	LO309E110	Dynamic mezzanine
310	LO310A001	LO310E115	Dynamic mezzanine
311	LO311A001	LO311E023	Dynamic mezzanine
312	LO312A001	LO312E036	Dynamic mezzanine

Aisle	First_Location	Last_Location	Type
313	LO313A004	LO313E024	Dynamic mezzanine
314	LO314A004	LO314E024	Dynamic mezzanine
315	LO315A001	LO315E007	Dynamic mezzanine
316	LO316A001	LO316E034	Dynamic mezzanine
401	LO401A001	LO401E057	Dynamic mezzanine
501	LO501A001	LO501E090	Dynamic mezzanine
BAY	LOBAY001	LOBAY005	Prepared support drop location to shipping
CDC	LOCDCA001	LOCDCA001	Prepared support drop location to control
CDP	LOCDPGP001	LOCDPGP002	Prepared support drop location to control
FIN	LOFINDEPREP1	LOFINDEPREP1	
QUA	LOQUAEXP001	LOQUAEXP001	Shipping dock
RE	LOREJET	LOREJETZPA	Reject location when ZPA storage rule (mezzanine) found no locations
REL	LORELAISZPA01	LORELAISZPA01	Relay location for partial replenishment in ZPA
TRA	LOTRANS001	LOTRANS001	Transitional location during preparation support cancellation

Zones

Zones play an essential role in logistics activity management.
They intervene at several levels:

- In storage rules
- In allocation rules
- In replenishment processes
- In ZPA transfers

- As well as in equipment operator mission management (access via radio)

When creating a new location, it is imperative to associate it with the appropriate zones. Incorrect association can lead to malfunctions in one or more of the mentioned processes.

Expected zone configuration:

🧩 **For proper allocation and replenishment functioning:**

- All locations: LO_ALL_PRE
- Reserve locations: LO_STD_STK_ALL (storage zone) + LO_STD (picking zone)
- Dynamic picking locations: LO_ZPA_LOREA + (LO_ZPA_MEZZ or LO_ZPA_RACK)
- Dedicated picking locations (all these zones):
    - LO_PICKING_RA01
    - LO_PICKING_RL01
    - LO_PICKING_RP01
    - LO_PICKING_RC01
    - LO_PICKING_RLSC
    - LO_PICKING_ALL

🗄️ **For proper storage functioning:**

- Reserve locations: zone of type LO_RACK_XXX
- Dedicated picking locations: N/A
- Dynamic picking locations: zone of type LO_P_XXXXX

🚜 **For equipment operator mission retrieval:**

- Dedicated picking locations: LO_CARISTE_DEDI
- Dynamic picking locations: LO_CARISTE_ZPA

ZONE	LABEL	ZONE TYPE	COMMENTS
LO_RACK_ALL	RACK_FULL_PALLET_RESERVE_ALL	1	Used in reserve storage rule 0001
LO_RACK_QLT	RACK_FULL_PALLET_RESERVE_QUALITY	1	Used in reserve storage rule 0001
LO_RACK_DMG	RACK_FULL_PALLET_RESERVE_DAMAGED	1	Used in reserve storage rule 0001
LO_CAGE_CPD	CAGED_RACK_FULL_PALLET_AEROSOLS_CPD	1	Used in reserve storage rule 0001
LO_RACK_CPD	RACK_FULL_PALLET_RESERVE_CPD	1	Used in reserve storage rule 0001
LO_RACK_PPD	RACK_FULL_PALLET_RESERVE_PPD_LPD	1	Used in reserve storage rule 0001
LO_CAGE_LDB	CAGED_RACK_FULL_PALLET_AEROSOLS_LDB_PPD_LPD	1	Used in reserve storage rule 0001

ZONE	LABEL	ZONE TYPE	COMMENTS
LO_RACK_LDB	RACK_FULL_PALLET_RESERVE_LDB	1	Used in reserve storage rule 0001
LO_TRANSITIONEL	Buffer zone for cycle count	9	Transitional location used when deleting a shipping support (canceling already prepared mission)
LO_P_CPD_AER	Picking_CPD_Aerosol	1	Used in ZPA storage rule 0002 and picking rule 0003, finer than global config
LO_P_ALL_AER	Picking_LDB_PPD_LPD_Aerosol	1	Used in ZPA storage rule 0002 and picking rule 0003, finer than global config
LO_P_PPD_ALL	Picking_PPD_ALL	1	Used in ZPA storage rule 0002 and

ZONE	LABEL	ZONE TYPE	COMMENTS
			picking rule 0003, finer than global config
LO_P_CPD_ALL	Picking_CPD_ALL	1	Used in ZPA storage rule 0002 and picking rule 0003, finer than global config
LO_P_LPD_PLV	Picking_LPD_PLV	1	Used in ZPA storage rule 0002 and picking rule 0003, finer than global config
LO_P_LDB_ALL	Picking_LDB_ALL	1	Used in ZPA storage rule 0002 and picking rule 0003, finer than global config
LO_P_LDB_PLV	Picking_LDB_PLV	1	Used in ZPA storage rule 0002 and picking rule 0003, finer

ZONE	LABEL	ZONE TYPE	COMMENTS
			than global config
LO_P_ALL_PLV	Picking_ALL_PLV	1	Used in ZPA storage rule 0002 and picking rule 0003, finer than global config
LO_P_CPD_MK	Picking_CPD_makeup	1	Used in ZPA storage rule 0002 and picking rule 0003, finer than global config
LO_P_PDD_ALL2	Picking_PPD_ALL_2	1	Used in ZPA storage rule 0002 and picking rule 0003, finer than global config (_2 indicates overflow zone)
LO_P_LPD_ALL	Picking_LPD_ALL	1	Used in ZPA storage rule 0002 and picking rule

ZONE	LABEL	ZONE TYPE	COMMENTS
			0003, finer than global config
LO_STD_STK_ALL	Storage zone all	1	Origin storage zone in ZPA global config
LO_STD	L'Oréal Reserve picking zone	2	Search zone used during full pallet allocation in launch parameters
LO_PICKING_ALL	L'OREAL_Picking_dedicated_ALL	3	Used in picking allocation rules in launch parameters. Must contain all dedicated picking locations
LO_ZPA_LOREA	L'Oréal ZPA picking zone	2	Used in ZPA global config (destination zone) and in allocation rules in launch parameters.

ZONE	LABEL	ZONE TYPE	COMMENTS
			Must contain all dynamic picking locations
LO_ZPA_MEZZ	L'Oréal ZPA mezzanine picking zone	2	Used in allocation rules to differentiate mezzanine picking from rack picking (rack picking grouped with dedicated picking missions)
LO_ZPA_RACK	L'Oréal ZPA rack picking zone	2	Used in allocation rules to differentiate mezzanine picking from rack picking (rack picking grouped with dedicated picking missions)
LO_ALL_PRE	ALL LOCATIONS RESERVE+ZPA+PICKING	2	Search zone used during availability

ZONE	LABEL	ZONE TYPE	COMMENTS
			control in launch parameters, must contain all locations
LO_PICKING_RL01	L'OREAL Picking dedicated replenishment RL01	3	Used in dedicated picking replenishment rules as only one quality code is configurable per zone
LO_PICKING_RC01	L'OREAL Picking dedicated replenishment RC01	3	Used in dedicated picking replenishment rules as only one quality code is configurable per zone
LO_PICKING_RA01	L'OREAL Picking dedicated replenishment RA01	3	Used in dedicated picking replenishment rules as only one quality code is

ZONE	LABEL	ZONE TYPE	COMMENTS
			configurable per zone
LO_PICKING_RP01	L'OREAL Picking dedicated replenishment RP01	3	Used in dedicated picking replenishment rules as only one quality code is configurable per zone
LO_CARISTE_DEDI	Equipment operator zone for fixed picking replenishment	8	Equipment operator zone to distinguish all equipment operator movements when entering RF module, copy of lo_picking_all
LO_CARISTE_RECP	Equipment operator zone from reception	8	Equipment operator zone to distinguish all equipment operator movements when entering RF module

ZONE	LABEL	ZONE TYPE	COMMENTS
LO_CARISTE_ZPA	Equipment operator zone for ZPA transfer	8	Equipment operator zone to distinguish all equipment operator movements when entering RF module, copy of LO_ZPA_LOREA

- **IV. Storage Rules**

There are three main storage rules, each with a specific role in flow management:

1. **Rule 00001 – Reserve location search**
   - Applied by default on items and during reception
   - Directs products to standard reserve locations
2. **Rule 00002 – Replenishment to picking (ZPA)**
   - Used to direct picking replenishments to appropriate mezzanine locations, particularly via ZPA transfer
   - Ensures proper distribution of reserve stock to dynamic picking zones
3. **Rule 00003 – Global picking storage**
   - Used for direct storage in picking zone, particularly from reception
   - Unlike rule 00002, it also includes storage in dedicated picking locations (often on pallet racks)

  - Includes a specific to control the maximum picking stock threshold configured on each item, to avoid exceeding defined capacity

Detail rule 0001: Reserve storage

The first four storage rules direct products to caged rack zones, dedicated to storage of dangerous products such as aerosols.
These rules are specific to each division, to ensure separation compliant with safety requirements.

The next four rules guide standard products of each division to their respective storage zones, according to typology and associated management mode.

Finally, the last two rules direct obsolete products or those with quality issues to dedicated closed zones, intended for this type of treatment.

Detail rule 0003: Picking storage

The first five rules concern items classified as "fixed picking".
They search for a location in fixed picking zones of type LO_PICKING_XXXX (where XXXX corresponds to the quality code).

The next eleven rules intervene to complete an existing location, based on:

- Available volume
- Various item criteria (aerosol level, PLV type, division, etc.)

Finally, the last eleven rules correspond to free searches.
They allow finding a new location in the same zones as before, with the same breakdowns (aerosols, PLV, division), but they additionally integrate the picking threshold defined in the item master data (2 or 5).

This threshold represents the maximum number of locations authorized in dynamic picking for the same item.

For more details on configuration of these rules, refer to document SPD 020.

Detail rule 0002: ZPA storage used in dynamic picking transfer

This rule contains sequences similar to rule 00002.

- First, a first sequence completes an existing location based on available volume
- The next sequence searches for a new location in the appropriate zone

- The main difference with previous free searches is the absence of picking threshold control: this rule being used in the context of replenishment, it must accept all location proposals
- Finally, a last control sequence detects failure cases (no applicable sequence). In this case, the ZPA transfer is not blocked: the support is redirected to a reject location, where it must be processed manually by the responsible party.

- **V. Replenishment Rules**

In the AGL – L'Oréal Morocco warehouse, several divisions coexist, each with different volumes and order typologies.

To respond to all preparation modes, picking layout is managed according to two distinct approaches:

1. **Dedicated picking (classic)**
   → Locations are fixed and reserved for specific items
2. **Dynamic picking**
   → Managed via transfer to Advanced Preparation Zone (ZPA), allowing flexible allocation of locations based on operational needs

Classic dedicated picking replenishment

A picking location—generally located at the first level of the pallet rack—is reserved for a specific item.
In this case, the item is classified as "fixed picking" and has a picking layout defined in its parameters.

Note that each fixed picking layout should always contain two lines:

- One with the standard quality code corresponding to the division
- A second with the RLSC quality code, associated with copacking activity

This configuration ensures that products attached to the RLSC quality code can also be replenished in the context of copacking order preparation.

This operation is standard in SPEED: replenishment is automatically triggered at order launch when the available quantity in the dedicated location is insufficient to cover the need.

Dedicated replenishment configuration:

- Replenishment is limited to the need of the current wave
- The replenishment handling unit corresponds to the complete support: the system will therefore seek an entire support in reserve, whether or not it corresponds to the pallet handling unit

Dynamic picking replenishment

Dynamic picking replenishment—i.e., on locations without fixed picking layout (when an empty location can be replaced by any other reference)—is not managed by standard in SPEED.

To respond to operational needs, we implemented the ZPA transfer, associated with some specifics.

Recall that ZPA transfer is an anticipated process that relies on the order portfolio to bring down the need to a specific picking zone, which will then be targeted during order launch.

Objectives for L'Oréal:

1. Bring down only unit needs, not full pallet needs, already covered by complete support picking 006 in launch parameters
2. Avoid complete support replenishment (standard SPEED) and allow choosing a partial quantity to bring down, to optimize space in dynamic picking, often heavily utilized

ZPA transfer configuration:

The ZPA transfer is configured according to this logic:

- Maximum handling unit to consider: CASE (to not take complete support need)
- Origin storage zone: all reserve stock
- Destination zone: LO_ZPA LOREA (dynamic locations, whether in mezzanine or pallet rack (level C))
- And finally targets only items whose classification is items/Type_Picking/Dynamic

ZPA transfer calculation is based on orders having a preparation date.
By default, no preparation date is sent down, which allows the scheduler to precisely target orders on which they wish to trigger ZPA transfer—in other words, only those for which a preparation date has been entered.

Support status management:

By default, ZPA transfer generates equipment operator transfers with status 03.
A specific was added to allow generation of replenishments with support at status 04.

Process harmonization:

Thanks to this evolution, the same picking replenishment module can now be used to execute ZPA transfers.
This module was also modified to provide operators with real-time information on the number of units needed in dynamic picking at time T (displayed on the last line of the screen).

Partial replenishment and stock visibility

This operation allows performing partial replenishment, adjusted to minimum need.
For example, if the need is 200 units, the system will bring down the entire support to cover the demand as in the example above. But if the quantity of need is less than the support, the operator has the choice to enter the minimum quantity. There is no control, however. If they enter less than the need, the system will accept.

A specific view on stock allows visually differentiating replenishment types if necessary:

- Orange flag: ZPA transfer
- Blue flag: dedicated replenishment

- **VI. Description of Different Reception Processes**

There are several types of ASNs (expected receipts), as mentioned previously.
The reception process is identical for INB, PPP and BOM ASNs.
From creation of the expected receipt by the interface, the system automatically generates the associated reception.
This reception initially contains no lines, as effective stock creation is performed via the "Multi BL Reception" radio module, renamed for the occasion "ASN Reception".

Generally:

- Reception of non-expected items is not authorized
- Reception of quantities greater than those expected is not authorized either

Regarding returns, the process differs:
there is no expected receipt transmitted by the client.
Choice of items and quantities to receive is therefore left to the discretion of the operator in charge of return processing.

Radio module multi BL reception (without expected receipt)

For return management, operators will be able to create a reception without expected receipt via the menu -> reception without expected receipt and the RETURN tab.
They can then receive any item and quantity. The applied quality code is RL98.
They generally use generic labels for creation of support identifiers.

Classic radio module multi BL reception

The reception module is configured to successively request:

1. The BL number
2. Then the support number

Several decoding mechanisms were implemented at this stage to allow interpretation of EAN128 labels containing:

- The SSCC of the support
- The item code
- And the lot number

The SSCC number is generally retained to identify complete supports in the system.

Reception-specific specifics:

Two SPDs modify radio behavior in reception:

- **SPD09**: Authorizes entry of complementary logistic data, such as:
  - Item dimensions
  - Weight
  - Expiration date tracking
  - Pallet and case handling units
- **SPD011**: Allows automatic decoding of manufacturing date from lot number

Regarding SPD09, distinguish two distinct data groups:

- **Item dimensions**: If at least one item dimension or weight is not entered, the dimension entry window will be displayed after creation of a first support
- **Traceability data**: Traceability will be requested only once, at first reception of the item as well as mandatory pallet and case handling units

Regarding SPD011, it triggers if the item is checked lot1 tracked. The system requests the lot number (if not recovered from the initially scanned EAN128).

Automatic calculation of manufacturing dates and expiration date (from product shelf life present in item master data and decoded manufacturing date).

For decoding rules of manufacturing date from lot, refer to SPD 011.

Reception closure

Reception is closed in admin.
Logistic closure is mandatory before proceeding to final closure.
SPD015 intervenes at this stage:
it automatically deletes expected receipt lines for which there remain non-received quantities.

Indeed, the remainder is not managed in SPEED, but directly in the host system (ERP SAP). If a remainder must be processed, a new expected receipt will be generated and sent down by SAP.

Important note:
A SAP constraint prevents reception of an expected receipt line at 0.
By default, the EDI interface enters 1 unit when the quantity to receive is zero.
A stock discrepancy is then corrected manually on both sides by operators.

Post-reception

Created supports are stored progressively by equipment operators, as operations proceed, via the equipment operator storage transaction.
They have the possibility to manually switch to rule 0003 to perform direct storage in picking zone, if necessary.
No allocation is possible from reception docks:
as long as merchandise is not physically stored and validated in stock, it remains non-allocatable.

- **VII. Description of Different Preparation Processes: Launch to Shipping**

General Information

Order launch is performed on the fly, orders rarely being sent down in advance.
Similarly, ZPA transfer is triggered just before each wave launch, to supply picking zones as close as possible to real need.
Several wave launches are performed throughout the day.
Replenishments are performed hot and prioritized dynamically based on immediate needs of preparers.

The flowchart below presents the steps of the preparation process from order launch to shipping.

Preparation Steps:

1. Selection of order batch to launch and modification of preparation date
2. Launch of availability calculation
3. Control of shortages and correction if possible (using stock view/allocated orders -> allocated and shortage orders) -> a specific view is used in the Stock screen, order in entry
4. Launch of ZPA transfer
5. Launch of orders
6. Printing of carton picking and reserve reports

Preparation:

Use of radio module: multi-order preparation

1. Choice of mission type
2. Scan of mission number displayed on picking report (one or several)
3. Beginning of picking process (scan location, scan item, then enter quantity)
4. If item 'AD code' tracked, scan as many AD numbers as units to pick for LPD or as cases for PPD
5. Scan of drop support (prepared support)
6. Deposit of supports in the finalization location at end of preparation

Finalization:

Each support is scanned via the finalization module, weight is entered and the support is finalized.
Once all supports are finalized, the order passes to packed status and SHP is sent.

The client then sends back an order update with the invoice number, payment method and shipping conditions.
Once this information is updated, the shipping report is printed by operators then attached to the load.

The client also sends an invoice by email which is printed and attached to shipping documents.

Note: The load creation process and support loading is not yet used by operators but was part of new needs identified in the functional analysis.

Allocations

Allocations are divided following this sequence:

1. Availability control on all stock, zone LO_ALL_PRE
2. Allocation priority on complete supports in reserve. This encompasses all items and is based on zone LO_STD
3. Stock service PST allocation on dynamic picking locations in rack LO_ZPA_RACK
4. Stock service PST allocation on dynamic picking locations in mezzanine LO_ZPA_MEZZ
5. 5 picking PPI allocations on dedicated picking locations for each quality code, including copacking on zone LO_PICKING_XXXX (XXXX = quality code, RC01 RL01 RA01 RP01 RLSC)
6. Stock service for copacking in RLSC on zone LO_ZPA_LOREA
7. Missing service to recover what passed availability control but was not allocated on any other sequence

Step	Description	Zone / Rule
1	Global availability control	LO_ALL_PRE

Step	Description	Zone / Rule
2	Complete support allocation in reserve	LO_STD
3	ZPA stock service (rack)	LO_ZPA_RACK
4	ZPA stock service (mezzanine)	LO_ZPA_MEZZ
5	Picking PPI (dedicated zones)	LO_PICKING_XXXX (RL01/RC01/RA01/RP01/RLSC)
6	Copacking stock service	LO_ZPA_LOREA
7	Missing service	Following availability control

We use the PICKING pattern for allocations 5 and 3 to be able to group them in the same missions as picking is done in the rack zone.

Missions

Generation and mission structure

Missions are generated and divided according to several criteria:

- Division
- Zone (mezzanine or rack)
- Number of mission lines (variable by division) for picking, or number of supports (maximum 10) for reserve picking
- Picking type
- Preparation order number (OPE_NOOE)

Replenishment missions

Replenishment missions are generated only for dedicated picking.

- Source search zone: LO_STD_STK_ALL
- Target picking zones: zones of type LO_PICKING_XXXX (dedicated picking)
  Missing mission lines can also be generated per order to ensure flow completeness.

Prepared supports

Prepared supports are created automatically from mission generation.
There is no pre-boxing:
one prepared support = one mission.

Mission nomenclature

MISSION TYPE	DESCRIPTION
PICK_CPD	Picking division CPD
PICK_LPD	Picking division LPD
PICK_PPD	Picking division PPD
PICK_LDB	Picking division LDB
PICK_CPD_M	Picking CPD mezzanine
PICK_LPD_M	Picking LPD mezzanine
PICK_PPD_M	Picking PPD mezzanine
PICK_LDB_M	Picking LDB mezzanine
PALLET_CPD	Complete support picking CPD
PALLET_LPD	Complete support picking LPD
PALLET_PPD	Complete support picking PPD
PALLET_LDB	Complete support picking LDB

M: mission performed in mezzanine

PALLET: complete support picking mission

- **VIII. Description of Interfaces and EDI Operation**

General file exchange operation

Exchanged files transit via L'Oréal's SFTP server:

- The OUTBOUND directory contains files deposited by L'Oréal for AGL

- The INBOUND directory contains files generated by AGL and retrieved by L'Oréal

Downstream flow (L'Oréal → SPEED)

1. Flat files (in text format, separated by tabs) are retrieved by the WebMethods service
2. WebMethods converts the flat file to a JSON format API call
3. This call is transmitted to BK System's LINKUP service, which:
    - Interprets the JSON
    - Then updates SPEED with received data

Return flow (SPEED → L'Oréal)

1. Similarly, WebMethods performs recurring calls to LINKUP APIs
2. As soon as data is available, LINKUP returns JSON content
3. WebMethods converts this JSON to a flat file (.txt), then deposits it on L'Oréal's SFTP in the corresponding directory

LINKUP marking management

LINKUP marking relies on the KEYU of the main table.
It allows identifying already transmitted data, to send them only once.
If needed, it is possible, via SpeedService, to delete the marking to make elements available again during the next API call.

INBOUND Flows

ASN

There are 3 types of ASN in the EDI flow:

- PPP → Local ASN from Morocco
- INB → International ASN
- BOM → ASN allowing creation of copack bundles

The difference between PPP and INB lies in the presence of an inbound number in the case of INB in addition to the Purchase order number. For PPP, both fields will have the same value.

Return and cancellation ASNs are outside EDI. They are created from Speed, see ASN return section.

BOM ASN specificities:

The BOM ASN sent by the client contains a header line representing the quantity of bundle to create. It serves to create the BOM ASN.

In the client file, BOM ASN lines represent components constituting the bundle. They serve to create DO COPACK.

ASN Return

There are two types of ASN return:

- Cancellation (with reference), partial or total
- Customer return without reference

Creation of return and cancellation expected receipts is not sent down by the client.
The process differs depending on return type.

Cancellation:

We implemented a specific internal EDI flow via linkup that allows creating a reception expected receipt matching the order being canceled (the reception is also created by interface).

To trigger it, operators simply check the cancellation_order_creation_expected_auto box and enter a cancellation_reason_code.

The return expected receipt is created with the invoice number as order reference. The flow runs approximately every 5 minutes.

Operators can then receive items as for a classic reception either totally (order entirely canceled) or partially. It is important to properly ship the reception to send back the good receipt message to L'Oréal.

Return:

Returns without reference are created directly from the radio. Using the multi BL reception module, then reception without expected receipt and finally the Return tab.

Similarly, ASN closure will trigger generation of the GR message.

An additional operation is nevertheless required in this case: enter the following complementary information (beginning with 'return' + receptionType) in the reception header.

If this information is missing (particularly receptionType), no message will be sent back.

Distribution Order

The distribution order flow is classic, orders are typed 'customer_order' in the complementary information OPE_ALPHA11 (unlike do Copack which are typed 'copack_order').

OUTBOUND Flows

In most transactional outbound flows, a unique WMSTransactionNumber is sent back.
The host system (SAP) does not allow integration of multiple messages with the same wmsTransactionNumber.
It is sometimes necessary to resend a message that was in error in SAP. In this case, a new WMSTransactionNumber must be used.

We therefore sometimes use in the JSON a forceReprocessingFlag. If this is at 1, then WEBMETHODS will modify the transaction number by adding a suffix XXXXXX_1 XXXXXX_2 XXXXXX_3 and so on, depending on the number of times the message is reprocessed.

This allows bypassing the SAP constraint that accepts only one unique WMSTransactionNumber per flow.

Good Receipt

The GR flow is triggered by complete closure of the expected receipt (logistic closure is not sufficient).

Good Receipt Return

The GR return flow is triggered by closure of the return ASN (cancellation or return).
Pay attention to properly enter complementary information in the case of RETURN (see ASN return section above).
In the case of cancellation, information is directly retrieved from the order during creation of the expected receipt and reception.

Shipment Confirmation

The preparation declaration is triggered when the order passes to status 060, packed.

Note: Order shipping does not trigger any flow.
However, it is not possible to ship an order as long as we have not received the doupdate (i.e., as long as the invoice number is not entered on the order's IC) -> SPD 026

Do Update

The DoUpdate flow is a specific. It is not present in Linkup. Files are saved on the server in:
H:\P-MAR-CA-S\ENVIRONNEMENTS\INTERFACE_MACASLEA1\COMMANDE_IMPORT

The DoUpdate flow updates the following information on the order:
BL comment / invoice number / netAmount / netAmountRet / ShippingType / DueDate / methodOfPayment

Inventory Movement

The stock adjustment flow only retrieves inter storage location transfers (or quality change in SPEED)
Additionally, a translation table is used to complete the message sent to the client. If the transfer is not present in the translation table, the message is not sent to the client.
Another particularity, if reason code 99 is used, the message is not sent to the client.

Translation Table 1:

Input Value	Output Value
RC01-RL99	RC01-RL99 [Transfer to Near Expiry]
RL01-RL96	RL01-RL96 [Damage]
RL01-RL88	RL01-RL88 [Transfer To Quality Issue]
RA01-RL97	RA01-RL97 [Transfer to Destruction]
RC01-RL96	RC01-RL96 [Damage]
RA01-RL99	RA01-RL99 [Transfer to Near Expiry]
RA01-RLSC	Transfer to BOM
RC01-RLSC	Transfer to BOM
RL98-RC01	RL98-RC01 [Return for Sales]
RL98-RL01	RL98-RL01 [Return for Sales]
RL98-RA01	RL98-RA01 [Return for Sales]
RL98-RP01	RL98-RP01 [Return for Sales]

Input Value	**Output Value**
RC01-RL98	RC01-RL98 [Return for Sales Reverse]
RL01-RL98	RL01-RL98 [Return for Sales Reverse]
RA01-RL98	RA01-RL98 [Return for Sales Reverse]
RP01-RL98	RP01-RL98 [Return for Sales Reverse]
RL98-RL97	RL98-RL97 [Return to Des]
RL97-RL98	RL97-RL98 [Return to Des Reverse]
RL98-RL96	RL98-RL99 [Return to Damage]
RL96-RL98	RL99-RL98 [Return to Damage Reverse]
RC01-RL92	RC01-RL92 [Transfer Missing]
RL01-RL92	RL01-RL92 [Transfer Missing]
RA01-RL92	RA01-RL92 [Transfer Missing]
RP01-RL92	RP01-RL92 [Transfer Missing]
RL92-RC01	RL92-RC01 [Reverse Missing]
RL92-RL01	RL92-RL01 [Reverse Missing]
RL92-RA01	RL92-RA01 [Reverse Missing]
RL92-RP01	RL92-RP01 [Reverse Missing]
RC01-RL95	RC01-RL95 [Transfer to Warehouse sales]
RL01-RL95	RLO1-RL95 [Transfer to Warehouse sales]
RA01-RL95	RA01-RL95 [Transfer to Warehouse sales]
RP01-RL95	RP01-RL95 [Transfer to Warehouse sales]
RL95-RC01	RL95-RC01 [Transfer to Warehouse sales-Reverse]

Input Value	Output Value
RL95-RL01	R195-RL01 [Transfer to Warehouse sales-Reverse]
RL95-RA01	RL95-RA01 [Transfer to Warehouse sales-Reverse]
RL95-RP01	RL95-RP01 [Transfer to Warehouse sales-Reverse]
RA01-RL96	RA01-RL96 [Damage]
RP01-RL96	RP01-RL96 [Damage]
RL96-RC01	RL96-RC01 [Damage-Reverse]
RL96-RL01	RL96-RL01 [Damage-Reverse]
RL96-RA01	RL96-RA01 [Damage-Reverse]
RL96-RP01	RL96-RP01 [Damage-Reverse]
RC01-RL97	RC01-RL97 [Transfer to Destruction]
RL01-RL97	RL01-RL97 [Transfer to Destruction]
RP01-RL97	RP01-RL97 [Transfer to Destruction]
RL97-RC01	RL97-RC01 [Transfer to Destruction-reverse]
RL97-RL01	RL97-RL01 [Transfer to Destruction-reverse]
RL97-RA01	RL97-RA01 [Transfer to Destruction-reverse]
RL97-RP01	RL97-RP01 [Transfer to Destruction-reverse]
RL01-RL99	RL01-RL99 [Transfer to Near Expiry]
RP01-RL99	RP01-RL99 [Transfer to Near Expiry]
RL99-RC01	RL99-RC01 [Transfer to Near Expiry - Reversal]
RL99-RL01	RL99-RL01 [Transfer to Near Expiry - Reversal]
RL99-RA01	RL99-RA01 [Transfer to Near Expiry - Reversal]

Input Value	Output Value
RL99-RP01	RL99-RP01 [Transfer to Near Expiry - Reversal]
RC01-RL88	RC01-RL88 [Transfer To Quality Issue]
RA01-RL88	RA01-RL88 [Transfer To Quality Issue]
RP01-RL88	RP01-RL88 [Transfer To Quality Issue]
RL88-RC01	RL88-RC01 [Transfer To Quality Issue Reverse]
RL88-RL01	RL88-RL01 [Transfer To Quality Issue Reverse]
RL88-RA01	RL88-RA01 [Transfer To Quality Issue Reverse]
RL88-RP01	RL88-RP01 [Transfer To Quality Issue Reverse]
RLSC-RLSC	Transfer to BOM
RL89-RL99	RL89-RL99 [Transfer To expired]
RL99-RL89	RL99-RL89 [Transfer To expired]

Inventory Snapshot
A stock snapshot is sent once per month.
Preview of file sent to client (item, client depot code, quality code, UN constant and quantity)

- **IX. Specifics**

1. Reception

- **SPD_009 – Item Reception: Entry of weight, dimensions, expiration date, LOT**
  This specific concerns the multi BL reception radio module. It allows entering dimensions and weight if missing on the item master. It also allows entering if the item has lot1 tracking as well as expiration date + pallet and case handling units only at first reception of the item.
- **SPD_011 – LOT Decoding Reception V5**
  This specific concerns the multi BL reception radio module. It allows decoding the

manufacturing date from the L'Oréal lot number and entering the expiration date based on the product shelf life configured on the item master.

- **SPD_015 – REA balance at reception closure V4**
  This specific concerns reception closure in ADMIN. It balances expected receipt lines linked to the reception that would not have been 100% received.

- **SPD_017 – REA_RFCV2 Control**
  This specific concerns creation of reception from expected receipt lines manually. It ensures that all lines with the same BL number have been selected. This spec is now obsolete as receptions are created directly by interface.

- **SPD_028 – IC inheritance between AEN expected receipts and REE receptions in admin**
  This specific concerns inheritance of ICs between AEN expected receipts and REE receptions which doesn't work by standard. Notably used for cancellation ASNs which require complementary information in header.

2. Storage & Replenishment

- **SPD_010 – Declarative Quality Item Storage V2**
  This specific concerns the equipment operator storage radio module. It allows configuring on an IC zones, quality codes for which it is not possible to store in the zone in question.

- **SPD_020 – Free search query for control of number of locations per zone vs Item IC**
  This specific concerns storage. It allows creating a free search with multiple criteria to filter items and search for locations in dynamic picking zones. It is notably used to respect the max dynamic picking threshold configured on the item master (IC).

- **SPD_024 – ZPA Storage – Transfer to replenishment (Client version)**
  This specific concerns ZPA transfer. It allows transforming the generated equipment operator movement into a replenishment movement. The picking replenishment radio module can then be used as for dedicated picking replenishments.

- **Spec_Reappro_Picking_Dynamique_ZPA**
  This specific concerns volume dynamic picking replenishment. It is being drafted by BK and not yet deployed.

3. Preparation, Control & Shipping

- **SPD_012 – Anti Diversion (AD CODE) V3**
  This specific concerns the multi-order preparation radio module. It allows requesting entry of the anti-diversion code (equivalent to serial number for fraud prevention) if the item is checked AD code tracking on its ICs. A control is operated on uniqueness of codes (unique on each unit for LPD and unique per case for PPD) and stores values in a specific SPE_AD table. This table is exported once per month to the client.

- **SPD_016 – Finalization Module**
  This specific concerns the finalization radio module. It allows printing shipping support label and generic label with counter.

- **SPD_026 – Order shipping blockage**
  This specific concerns order shipping in admin. We block shipping of customer_orders if there is no invoice number entered. This spec does not concern copack_orders.

- **SPD_027 – Prepared Support Control Module / Lot Control**
  This specific concerns the support control module. It allows displaying lots present on the controlled support and validating them or not. If not validated, the support is flagged control KO.

- **SPD_031 – Prepared support storage to loading dock associated with the order**
  This specific concerns equipment operator storage. It allows guiding a prepared support to a shipping dock that was configured at load level.

- **SPD_032 – Mission preparation path**
  This specific concerns sequencing of mission lines at the radio in the multi cmd preparation module. By default, the system sorts by zone then NOOR (sequence number). Dynamic picking and dedicated rack locations belonging to different zones, we had to remove the first sort on zone. We keep only the sort on NOOR.

4. Inventory

- **SPD_014 – Cycle count: Action to limit by User V2**
  This specific concerns the cycle count radio module. It allows limiting certain stock modification actions to certain user profiles only.

- **SPD_023 – AGL inventory with multiple counts 1.1**
  This specific concerns the inventory module.

- **X. Appendix**

Shipping Declaration

Carton Picking Report

Carton Reserve Picking Report

ASN Reception

The first column allows indicating if it is the first reception of the item.

End of Document